

JavaScript Task: Number Guessing Game

Objective

Create a Number Guessing Game using Conditions, Loops, and Random Numbers in JavaScript.

Requirements

1. Generate a random number between 1 and 100.
2. Ask the user to guess the number using `prompt()`.
3. Use a loop to keep asking until the user guesses correctly or reaches the maximum attempts.
4. Use if-else conditions to check the guess:
 - If the guessed number is greater than the random number, display "Too High!"
 - If the guessed number is less than the random number, display "Too Low!"
 - If the guessed number is correct, display "Congratulations! You Win!"
5. Count the total number of attempts made by the user.
6. Display the total attempts after the user wins.
7. Validate user input:
 - If the value is outside the range 1–100, show an appropriate message.
 - If the value is invalid, ask the user to enter a valid number.
8. Allow the user only 10 attempts to guess the number.
9. Display "Game Over" if the user fails to guess the number within 10 attempts.
10. After the game ends, ask the user whether they want to Play Again.
11. If the user chooses Yes, start a new game with a new random number.
12. If the user chooses No, display "Thank You for Playing!" and end the program.

Concepts to Use

- Variables (`let`, `const`)
- `Math.random()`
- `Math.floor()`
- `while` or `do...while` loop
- Nested Loops
- `if...else if...else`
- `prompt()`
- `alert()`
- Counter Variable
- Boolean Flag